

SHAOFU WANG

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PROFESSIONAL SUMMARY

Mechanical Engineering (Co-op) student at the University of Waterloo with professional experience in aerospace regulation and structural optimization at COMAC. Proven track record in reducing component mass by 40% while ensuring compliance with CCAR-25 aviation safety standards and rigorous Design for Manufacturability (DFM) principles. Adept at bridging conceptual design and physical production through technical documentation, cross-functional collaboration, and advanced 3D modeling.

TECHNICAL SKILLS

Design & Modeling	CATIA V5, SolidWorks, AutoCAD, GD&T (ASME Y14.5), DFM/DFA
Computation & Simulation	Ansys (Structural & Topology), Star-CCM+ (CFD), MATLAB/Simulink
Manufacturing & Fabrication	Laser Cutting, 3D Printing (SLA/FDM), Lathe/Mill, BOM Management
Programming & Data	Python (NumPy, Matplotlib), C++, Arduino/ESP32, Data Analysis

EDUCATION

University of Waterloo, Waterloo, ON *Expected Apr 2030*
Degree: Bachelor of Applied Science (B.A.Sc.) in Mechanical Engineering (Co-op)
Relevant Coursework: Mechanical Engineering Practice, Mechanics, Calculus, Linear Algebra, Chemistry for Engineers

WORK EXPERIENCE

Design Engineering Intern, Commercial Aircraft Corporation of China (COMAC) *Jan 2026 – Apr 2026*

- Structural Optimization:** Re-engineered a pilot's tray table assembly using Ansys Topology Optimization, achieving a 40% mass reduction (2.5 kg to 1.5 kg) while maintaining structural rigidity.
- Regulatory Compliance:** Validated design integrity through linear and non-linear FEA to meet CCAR-25 static load requirements, ensuring a 1.5 safety factor under extreme emergency landing conditions.
- DFM & Collaboration:** Partnered with manufacturing vendors to conduct Design for Manufacturability assessments, refining part geometries to eliminate complex undercuts and reduce CNC machining time.
- Electromechanical Design:** Prototyped a smart cabin overhead storage system featuring remote security locks driven by micro solenoid valves; authored comprehensive technical specifications for cross-functional review.
- Supply Chain Support:** Managed technical requirement documentation for a second-source procurement initiative, evaluating vendor capabilities to mitigate risks in the electric window blind supply chain.

ENGINEERING PROJECTS

"Launch & Snatch" Tabletop Arcade Toy *Fall 2025*

- Mechanism Synthesis:** Engineered a crank-slotted lever system; performed kinematic analysis in SolidWorks to ensure smooth 180-degree motion within a strict \$127.61 BOM budget.
- Problem Solving:** Resolved kinetic binding issues in the guide rails by implementing a pulley-assisted counter-moment design, significantly reducing friction torque observed during hardware testing.
- Simulation Validation:** Correlated Ansys Discovery stress results with physical load cell data, achieving a 92% correlation rate and proving the reliability of the virtual prototype.

Air-Bubble Drag Reduction for Ship Hulls *2024 – 2025*

- Hydrodynamic Testing:** Engineered a ventilated cavity structure on a 3D-printed hull model to evaluate the impact of air-bubble injection on skin friction reduction.
- Data Analysis:** Developed a MATLAB script using the least squares method for polynomial curve fitting, quantifying the inverse relationship between drag reduction efficiency and relative fluid velocity.

AWARDS & ACHIEVEMENTS

First Prize & Special Prize, Shanghai Youth Science & Innovation Fair *Mar 2024*
Third Prize, Shanghai Science Education Innovation Award *Mar 2024*